

SewSimple: An E-textile Prototyping Kit for Computational Making with BBC micro:bit

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Abstract

In this paper we present the “SewSimple”, an E-textile prototyping kit designed to support computational making with the BBC micro:bit, and discuss findings from a preliminary mixed-method evaluation involving twelve schoolchildren aged 9-11 at a local fee-paying primary school. Our research indicates that the “SewSimple” design effectively improves the construction of stable circuitry in E-textile projects compared to using the BBC micro:bit alone. Through qualitative analysis of interviews, photos, and video clips, we propose design recommendations that include hardware improvements such as repositioning grommets (plated-through-holes), customizing covers, and improving labeling. Additionally, we suggest enhancements in instructional design targeting crafting skills and conductivity knowledge, aimed at collectively enhancing user experience.

CCS Concepts

• **Human-centered computing** → **Field studies; Interaction design theory, concepts and paradigms**; • **Social and professional topics** → **Children**; • **Applied computing** → *Collaborative learning*; • **Hardware** → **Printed circuit boards**.

Keywords

Computational Making, E-textile, BBC micro:bit, Creativity, Gender, Design

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1 Introduction

Computational making [41] highlights the importance of making skills to complement computational thinking. E-textiles, in particular, have been shown to encourage children of all genders to develop an interest in Science, Technology, Engineering, Maths (STEM) [9, 46]. Previously, Rode et al. [38] argue that approximately £75 million have been spent on deploying the BBC micro:bit and “we have yet to find a single study that looked at the gender equity of the micro:bit as part of its development process...Given the importance of gender equity in STEM, this is a shocking oversight.” Rode and colleagues further argue that the BBC micro:bit is ill-suited for e-textiles due to the difficulty of *securing* conductive thread to it (as discussed in more detail below), and they recommend the development of a sewable edge connector to alleviate this problem citing its importance for ensuring the BBC micro:bit’s diverse use and promotes gender equity. To date over 10 million BBC micro:bits have been distributed and many of which have been given to low-income children or schools worldwide [32]. It is also the dominant computing teaching tool in the UK as one micro:bit was given for

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free to schools that opted into the program [31]. Thus, this is a significant sunk cost, and even if the BBC micro:bit is not the best e-textile platform out there, its integration into educational programs worldwide means that there is a need to better support its use for e-textiles to support gender equity. Thus, in this paper, we take up this challenge and introduce the “SewSimple” e-textile prototyping kit designed to address these challenges and enhance the BBC micro:bit’s potential to broaden gender participation in STEM. This late-breaking work’s contribution includes our early-stage prototype of “SewSimple” sewable edge connector, along with a preliminary mixed-method user study that provides suggestions for its refinement and usage. Our study demonstrates that the “SewSimple” kit facilitates computational making of creative and functional e-textile artifacts in a wearable format, effectively addressing the difficulty of creating stable circuits highlighted in prior work.

2 Related Work

Existing HCI research has explored technologies to teach computational thinking [47] and making [41] to young people (e.g. [16, 28, 29, 48]), increasingly with a focus on e-textiles [9, 17, 23, 24, 41, 43, 46]. In her seminal publication on the LilyPad Arduino, Buechley et al. [9] argued that the e-textiles domain specifically holds more potential for encouraging individuals from underrepresented genders to pursue STEM. E-textiles are commonly used to teach computing and creative design skills, requiring one’s incorporating digital and electronic features into fabric-based creations [34] and have gained particular prominence in computational making. While numerous e-textile platforms and construction kits exist [34], the LilyPad Arduino remains the most well-known. Its circular shape, compared to the square shape of the BBC micro:bit, allows for easy stabilization on fabric. Posch et al. [34]’s review highlights that this becomes a common design feature for e-textile tools and other commercial products such as Adafruit Gemma [1] and Flora [2] have adopted it for industry design. Likewise, this design decision inspired our “SewSimple” creation.

Sharing similar educational goals as the LilyPad Arduino, the BBC micro:bit, a pocket-sized programmable microcomputer intended to teach electronics, computational making, and programming to schoolchildren, was launched in 2016 with significant support from the British government with a “*particular focus on girls and those from disadvantaged groups*” [30]. However, it has been critiqued for insufficiently studying gender equity in its development and deployment [38]. While the latest BBC micro:bit research has particularly emphasized designing compatible systems to support its making ecosystems [3, 18], little prior work investigates its use as an e-textile platform [17, 19, 27, 38, 39]. Both Klimová [27] and Fey et al. [17] explore children’s creation of wearables using the BBC micro:bit but include no discussion of construction challenges, despite these being previously identified as a key challenge for children’s computational making [41]. It is important to note that children’s final e-textile projects presented in this paper appeared well-constructed with tidier stitches than the last author observed from previous e-textile workshops (e.g. [19, 38, 39]).

Rode et al. [38] detailed numerous challenges children faced when constructing stable e-textile circuits with the BBC micro:bit.

In addition to a host of other issues with children failing to understand conductivity, programming, or electronics, they specifically noted that children struggled to connect to the traces on the bottom of the BBC micro:bit. Rode and her colleagues identified three major problems in relation to hardware design and making skills. First, the board was not securely attached to the garment when children started sewing. Secondly, children had difficulty creating a solid electrical connection with the alligator clips, resulting in flickering sewable LEDs; occasionally, the alligator clips rotated and short-circuited the board. Third, to circumvent the use of alligator clips, children sometimes would sew the board directly to the garment to avoid the trouble using alligator clips but consequently would have to destroy their e-textile projects to reuse the board. To support easy construction experiences, prior work has designed construction kits using copper and stickers [35, 36] and snaps embedded in pre-cut fabric pieces [10, 21, 33]. However, these approaches do not directly address issues caused by the design of the BBC micro:bit. Recent work by Rode et al. [38] proposed a sewable edge connector that features soft clips and two additional sewing pins, which allows children to first experiment with sewing and then easily attach and remove the BBC micro:bit from fabrics. However, their proposed edge connector was rigid and less suitable for e-textiles as it failed to conform to the shape of the textile or the body. There has been considerable research into using soft printed circuit board (PCB) for e-textiles [8, 9, 20, 22, 50, 51] which highlight their material advantages that can adapt to the body and the garment for textile applications. While several edge connectors created for the BBC micro:bit are conceivably sewable, our expert review of them showed that they did not truly afford solid connections and utilized rigid PCBs [25, 26]. Identifying this gap in the research, our project aims to develop a flexible PCB edge connector that facilitates the BBC micro:bit’s use in e-textile design and computational making.

3 Method

3.1 “SewSimple” E-textile Prototyping Kit

The project’s main design goals were to address aforementioned construction issues related to attaching the BBC micro:bit to e-textiles [38]. We developed a research prototype called “SewSimple” which is made using soft PCB design, measuring 80mm by 80mm with a thickness of less than 1mm. With its flexibility and improved sewability, our design serves as an edge connector that enhances the utility and usability of BBC micro:bit in e-textile design and hopes to support more flexible and creative applications particularly in wearable fashion. Our e-textile prototyping kit (see Figure 1) includes one “SewSimple” soft PCB, one BBC micro:bit, one compatible battery holder, two AA batteries, one USB cable, a pack of sewing needles, a spool of conductive thread, a pack of sewable LEDs, alligator clips for rapid circuit building and testing, and five compatible plastic screws to mount the micro:bit to the board. We also provided print-outs of instructions that include various sewing and debugging techniques, along with information on building serial and parallel circuits and example codes for sewable LEDs as supplementary materials for new users.

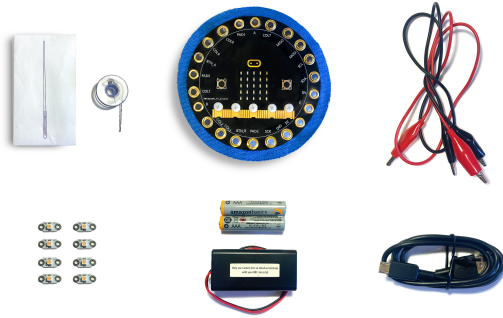


Figure 1: Hardware and tools included in the “SewSimple” kit.

3.2 Participants & Recruitment

We recruited an entire class of 12 children aged 9-11 through a research collaboration with a local fee-paying primary school in London, UK (see Appendix 8.1 for detailed participant information). Informed by best practices for recording gender information in HCI [44] and its application to recent child research [40], we asked our participants to self-describe their gender and provided three options: *boy*, *girl*, and *prefer not to say*. Our study involved five girls, seven boys, and zero “prefer not to say”. We received ethical approval from the universities’ ethics committees (#REC2071 & #CCI_CRESC_PhD_3) and obtained guardians’ informed consent. They were also invited to observe the second workshop as part of the school’s open-day program. Additionally, we obtained the children’s assent before the first workshop.

3.3 Procedure

Our study comprised three short workshops held at the school. Prior to the first session, children had had a one-hour lesson on BBC micro:bit programming and had assembled a simple e-textile with a soft switch lighting up an LED during their regular computing class. During the workshops, children worked in mixed-ability group pairs and further developed skills necessary for crafting e-textiles and programming the BBC micro:bits and additional electronics (e.g. sewable LEDs) and learned how to use our “SewSimple” kit (we utilized the BBC micro:bit V2 as they are already available and used for teaching at the school). Our instructional materials were co-designed with the school’s computing & maths teacher to ensure age-appropriateness and curriculum relevance. Informed by scenario-based design [42], children were given a design brief to create a smart t-shirt using the “SewSimple” prototyping kit with a scenario that was meant to help a new student “Ming” connect with her classmates. The student’s interests were provided to inspire design including favorite subject, animal, sport, game, etc. Selected workshop materials are presented in Appendix 8.2.

3.4 Data Collection & Analysis

The school teacher and the last author led the workshop instruction, with the other five researchers helping children and facilitating data collection including 1) photos of interim and final projects, circuit diagrams and screenshots of the final code, 2) video recordings, and 3) six fifteen-minute semi-structured interviews in pairs. During the three workshops, we collected 5 hours of video from the main-making phase (and we did not record time spent on instruction or decorative stitching). Our interview approach was selected to “*encourage(s) interaction between the group members rather than just responding to the adult’s questions*” [15, p. 433]. For our analysis, we employed reflexive thematic analysis (RTA) [4–7] for inductive coding for the interviews. Concurrently, the last author applied Critical Incident Analysis (CIA) [13] to analyze technical and non-technical issues around “SewSimple” usage from video clips and photos. We also deployed a post-workshop questionnaire using the Smileyometer 5 Point Likert Scale [37] to collect opinions on our early-stage prototype and general workshop experiences and performed descriptive and inferential statistical analysis such as Mann-Whitney U test. Results with p-values below the predetermined threshold ($p < .05$) were considered significant in our work. However, considering the limitations of our sample size, our discussions primarily emphasized qualitative findings.

4 Preliminary Results

Overall all six groups successfully created an interactive garment using the “SewSimple” prototyping kit and incorporated it into a t-shirt with decorative designs (see Figure 3). They utilized the buttons on the BBC micro:bit to activate the onboard LED display and 2-3 external sewable LEDs. Group 1 initially intended to trigger the external LEDs through conductive fabric and had a working prototype using alligator clips. However, they aborted this idea when they neglected to consult their circuit diagram while stitching. Groups 1, 2, 3 & 5 connected all their sewable LEDs securely so they stayed lit solidly. Group 4 connected two LEDs as per their circuit diagram, but one of them performed intermittently due to poor stitching. Group 6 required our research team’s help to iron on insulating patches, but ultimately was able to create stable connections to all of their sewable LEDs. In comparison with our previous computational making workshops with children, we observed markedly fewer bad traces sewn between the LEDs and the micro:bit; and using our “SewSimple” kit caused fewer flickering LEDs than using the micro:bit alone.

4.1 Qualitative Findings

4.1.1 Materiality of “SewSimple”. Children had different opinions about the material characteristics of “SewSimple”. Some children (P4, P5 & P7) found the board’s pliability made it easier for them to sew it onto textiles: “*It’s really, like soft. So it could, like, be easier to sew in micro:bits like or anything into, like clothes*”(P7). P4 reported that the circular shape allowed them to more securely attach the board to fabric: “*It’s fabric. It’s [a] good thing so it can hold it into place.*” and prompted design ideation to complete a more integrated product (“*...(It looks) like a whole earth circle...little metal circle and gold circles...It very inspires me as basically something, something with little circles in it can make something huge*”. In contrast, P1

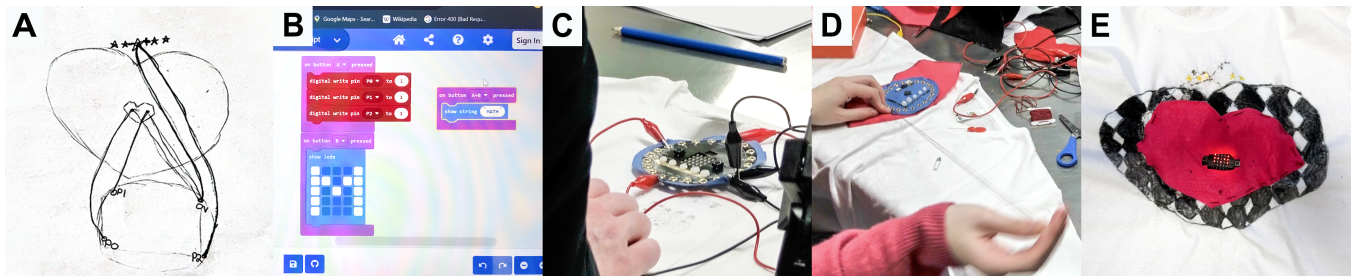


Figure 2: Children used our “SewSimple” kit in the workshop.



Figure 3: An overview of Group 1-6 creative design outcomes.

complained that the softness and the irritating sound of the plastic bending made it feel “flimsy”, raising concerns about it breaking easily during use. These differing perceptions highlight the need for further consideration of the material’s durability and ease of use.

4.1.2 Hardware Interface Design. Our hardware design features 19 pin-out holes and three sewing holes (we refer to all as grommets in light of the textile design context). We identified four issues related to these grommets.

“Too Many” Grommets. Our interview data revealed that upon initial viewing the “SewSimple” board, several children (P5, P8, P12) felt overwhelmed by the presence of 23 grommets and described

their immediate impressions such as “complicated” (P5, P12), “sophisticated” (P8) and expressed concerns about the requirement of “a lot of stitching” (P8) to attach the board to other materials. To overcome fears among novice users, it is recommended to offer a simplified version of the board with just 8 grommets; 3 for connections and 5 for programming (corresponding to the five plated-through-holes on the BBC micro:bit). This design mirrors the LilyPad Arduino which has two variants with different numbers of plated-through-holes for beginner and expert users (the ATmega328 with 22 and the Simple with only 11).

Confusion about Grommet Types. Our current design includes three grommets (A, B, & C) which are intended to securely sew the board to the fabric, whereas the other grommets had traces

connecting them to the micro:bit's pins and could be programmed. However, children did not initially appreciate the necessity of securely attaching the board, and thus not everyone used them. Only Group 4 & 6 used them unprompted, but they only connected grommet A. Group 3 used them only after being reminded by an adult following their query about what to do first. The rest of the groups skipped this step entirely. Specifically, Group 5 neglected to use the grommets completely and sewed a running stitch around the entire board, which made it impossible to see the labels and insert the battery. Thus, these two types of grommets must be clearly differentiated in the redesign.

Grommet Location. Our grommets were positioned further from the board edge than those on the LilyPad, partly to prevent children from pulling them out of the fabric lining (the back cover). However, some children (Group 2) pulled the edge tightly against the grommet to secure their stitches, causing the board not to lay flat against the textile they sewed onto. In contrast, other groups (4 & 6) did not pull it tightly and created considerably looser stitches as a result. These suggest that the grommets should be relocated to the edge of the board and that a ripstop material must be selected to prevent the board from tearing.

Label Visibility. Children intentionally concealed the “SewSimple” board and the micro:bit behind their textile designs (e.g. football, cat face/body, giant heart) tied into their project theme (Group 1, 3, 4, 5, 6) or whereas Group 2 hid it behind a pink square panel. For example, Group 1 cut and glued a piece of black felt to the circuit board to resemble a football, which entirely obscured the pin-out labels but allowed for them to put white spots on their ball. Similarly, Group 3, 4, & 6 initially sewed the board behind a cover, resulting in obscured labels; thus we helped add additional handwritten labels on the back to help identify programmable grommets. P7 from Group 6 said, “(because of) not having the numbers on both sides...we had to keep on tiling it over and writing the numbers onto the SewSimple so then we could find, like, where we had sew into...”. While Groups 2 and 5 did not incorporate the ‘SewSimple’ kit into their garment, they still utilized a colored felt to mask the board and the micro:bit. These observations suggest that our final design must include removable covers in a variety of colors, with labels clearly displayed on both the front and back.

4.1.3 Making Skills and Classroom Knowledge. After discussing with the teacher and the last author who are experienced educators working with young people, we also looked into non-technical issues impacting the use of the kit and highlighted two critical areas related to making and classroom knowledge.

Lack of Hands-on Skills. Our young participants were observed struggling with using scissors, threading needles and assembling the bolts and nuts with screwdrivers. Consequently, most children found it challenging to attach the micro:bit with the five screws, despite the screws being largely hand-tightenable. During the debriefing session, children reported that this task was hard: “they’re really hard to, like, get all your nails [screws] steady and you never know if they’re a proper type” (P7). These observations suggest that an alternative attachment method should be considered. However, snaps may not be an optimal solution as the last author has observed that students in their classes sheer the snap from the LilyPad Arduino SimpleSnap board.

Misunderstanding Conductivity in Circuit Construction. Previous e-textile research shows that children struggle to remember that their circuits can short out if conductive thread inadvertently touches any electrical components [11, 41]. In our study, although all groups completed a valid circuit diagram laid out on a plane and tested conductivity with electronics using alligator clips before moving onto sewing, challenges persisted during the “constructing” stage [41]. We observed the usual crossed traces of conductive threads when the guidelines [11] were inadvertently ignored.

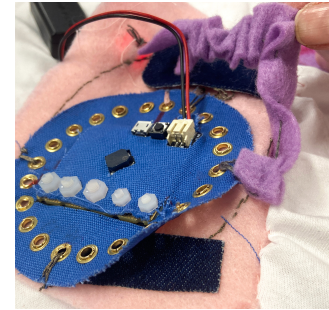


Figure 4: Group 6 misused grommets.

For example, Group 6 planned to sew through the unused grommets to connect the external LED to the grommet specified in their code. Their original circuit diagram included the whole t-shirt, with connections extending beyond the cat design. However, they attempted to replicate this using only the area of their cat design because they were focused on hiding the board first, which was barely the size of the micro:bit and was not attached to the t-shirt. We helped them fix this by insulating the conductive thread, enabling it to lay across the board without the risk of short-circuiting (see Figure 4). Similarly, Group 1 did not realize that much of the soft PCB was conductive and attempted to sew through the board to attach the spots to their soccer ball. As we will discuss below, these issues can be potentially addressed through instruction and rather than hardware redesign given that the material properties [39] remain unclear to the children.

4.2 Post-workshop Questionnaire

The workshop training (Q1), programming task (Q3), and general e-textile design experience (Q5) received positive scores, with averages of 3.67, 3.92, and 4.50 respectively (see Table 1). In contrast, design (Q2) and crafting tasks (Q4) such as sewing and circuit design were rated more neutrally, with average scores of 2.91 and 3.08. This shows that while the programming and the general experience were favorably received, participants expressed more reservations about sewing and circuit design tasks required for computational making. Furthermore, we assessed self-efficacy by asking participants to rate their nervousness versus confidence (Q6), yielding an average score of 3.50/5.00. Notably, there was considerable variability in participants’ feedback and high standard deviations. This reflects a diverse range of responses, potentially influenced by factors such as prior experience with electronics, comfort with completing tasks within a short timeframe, and skill levels required for sewing. Lastly, we asked the children if they wished to keep working on the project

Table 1: Our paper questionnaire was created using Read and MacFarlane [37]’s Smileyometer Likert Scales to assess participants’ general workshop experience and usability of the “SewSimple” prototyping kit. To mitigate biases caused by anchoring effects [14, 45], we designed our survey with 1 being the most negative and 5 being the most positive.

Question	Response Anchors	Mean	SD
Q1. Was it hard or easy to understand the training session?	Very Hard(1) → Very Easy(5)	3.67	0.65
Q2. Was sewing the prototype hard or easy?	Very Hard(1) → Very Easy(5)	2.91	1.00
Q3. Was programming the micro:bit hard or easy?	Very Hard(1) → Very Easy(5)	3.92	1.24
Q4. Was designing the circuit hard or easy?	Very Hard(1) → Very Easy(5)	3.08	1.00
Q5. Was it boring or fun to design your project?	Very Boring(1) → Very Fun(5)	4.50	0.67
Q6. Were you nervous or confident making your project?	Very Nervous(1) → Very Confident(5)	3.50	0.67

(11/12 “Yes”, 1/12 “No”), and if they would recommend “SewSimple” kit for others to use (12/12 “Yes”). The one participant (P1) who did not want to keep working was from Group 1, which faced many programming challenges for their project. This group included P2 who was “very confident” about their programming (5/5) and confident about their circuit-creating ability (4/5) and proposed the most complex design using conductive fabrics as a switch to light up the LEDs. Conversely, P1 reported low self-efficacy (2/5).

In addition, we ran Mann-Whitney U test and found gender statistically significant difference in responses of Q1 ($U = 29.5$, $p = .038$), Q3 ($U = 31.0$, $p = .026$), and Q6 ($U = 29.5$, $p = .033$), although with such a small sample size this finding is weak. In the statistically significant cases, boys demonstrated greater confidence in processing the training materials, performing programming tasks, and completing the project. Our findings align with prior research on gendered technology experiences with children [12], calling for further research to investigate non-technical factors such as social and cultural norms that might influence gender attitudes towards new technology and providing targeted interventions to improve girls’ computational interest and digital experiences. Overall, given that the post-questionnaire sample consisted of only 12 participants (five girls, seven boys, and zero “prefer not to say”), limiting the statistical power of these findings, additional research is required with a larger sample size to investigate these findings further. Despite this, all task ratings are still skewed towards neutral to positive end, aligning with our qualitative data and indicating an overall good user experience with the “SewSimple” kit.

5 Discussion & Design Implications

5.1 Hardware Improvements

Our preliminary evaluation leads us to make four key suggestions for improving the hardware usability.

- (1) **Beginner and Expert Version:** Akin to the LilyPad Arduino ecosystem, we recommend a basic version of the “SewSimple” board with fewer pins to complement this full-function board.
- (2) **Differentiate Grommets Function:** Based on our observation that children could not differentiate between the grommets for creating a solid sewn connection and those for programming, we recommend using two distinct colors for

the grommets. This will help distinguish the grommets for attachment from those for programming and clear instructions should be provided to guide novice users.

- (3) **Reposition Grommets:** To avoid when looping threads through grommets that sometimes can cause materials to buckle or knots to loosen for novice users, we recommend repositioning the grommets closer to the edge and using a ripstop fabric, keeping in mind children’s views on materiality.
- (4) **Customization Covers & Double-sided Labels:** Given that children wanted to obscure the board, a variety of removable covers should be available. Further, the board should be clearly labeled on both the front and back.

5.2 Instructional Design

Prior research highlights the effective role of integrating instructional design with usability to enhance user experience [49]. In our study, while some issues with the “SewSimple” can be addressed through hardware improvements, others can be addressed through instructional design practices. For instance, a rigid circuit board would prevent children from sewing through it, but it is vital that we use a soft circuit board to ensure e-textile garments are comfortable to wear, thus we must instead add training material that children should not sew through the board. Children did not understand the PCB had electrical traces connecting the grommets to the BBC micro:bit, and children need to be educated about how circuit boards are constructed. Thus, we identified five other areas where the instructional materials included as part of the “SewSimple” kit must be improved:

- (1) **Start with Sewing Pins:** Our instruction should emphasize securing the “SewSimple” to the material first using grommets A,B, and C. Children without electronics experience do not value the importance of this for creating a stable circuit as they lack the experience, thus insisting on it is the only option.
- (2) **Unobstructed Pin Labels:** Pin labels must stay visible during the design and testing phase to facilitate debugging. In doing so users or helpers can check their circuit functions.
- (3) **Pouch for Battery:** Experienced tutors should instruct novice users to include a pouch for the design stage to both support the battery and make it accessible. As the placement of the battery pack was not part of the initial design kit, batteries

were left dangling from the garment, which over time would cause the batteries to fail (the weight of the battery pack can pull at the connect and break).

- (4) **Reminding to Use Circuit Diagram:** Our instruction should guide children to complete and test a circuit diagram before other decorative design activities. Unfortunately, when the children started stitching, they ignored their plan, as outlined above this resulted in stitching their circuits in ways that shorted it. It is imperative to remind children to follow their diagrams while stitching.
- (5) **Step-by-Step Making:** Instructors should clarify steps and logic to complete relevant tasks for e-textile design. Our young participants were more preoccupied with the aesthetics of the T-shirt and spent far too much time cutting and gluing felt. Gluing of additional parts should be left to the end of the activity once the final assembled circuit has been tested and verified to function. This caused the material to get wet so the children were not able to continue with their project until it was dry, and risked shorting their circuit while wet.

Ultimately, these UX issues are easily addressed with improved instructions and do not require hardware changes.

6 Limitations and Future Work

Given the BBC micro:bit's prominence in education [31, 32], enhancing its capabilities to accommodate diverse STEM activities is essential, particularly in the domain of e-textiles, a creative engineering practice known for its potential to improve gender equity [23], even if the micro:bit is not currently the best platform for e-textile applications. Thus, our small-scale user study revealed several issues related to hardware design features and provided insights for improving the surrounding educational design in our kit. Our findings are adequate for the next iteration of the hardware. However, at that point, a larger longitudinal study including a wider age and skill range is required to validate the hardware prototype and ensure our educational objectives regarding computational making are met.

As an early-stage prototype evaluation, this work points out new opportunities for both hardware and instructional design. These include providing two variants of the “SewSimple” board for novice and expert users, addressing issues with grommet location, improved front and back labeling, and better differentiation of sewing and programmable grommets, and including cover options as part of the kit. We also intend to explore replacing the screws which were fiddly for small hands with more robust and user-friendly methods for attachment and detachment from textiles. Finally, by collaborating with educators and learning design specialists and running additional user studies, we will create better instructional materials for the “SewSimple” kit. Our long-term goal is not a research prototype, but being able to construct a product that will widely go into production encouraging gender equity in STEM. Making these hardware improvements, combined with instructional design, would allow us to arrive at a robust usable product.

7 Conclusion

In this research, we present the “SewSimple” E-textile prototyping kit aimed at supporting the use of BBC micro:bit in e-textile design, particularly to foster computational-making skills and technical literacy through creating expressive and fun wearable projects. Our preliminary evaluation from a mixed-method “in the wild” user study demonstrates interest in and positive experiences with our kit. Notably, it addresses previously discussed challenges assembling functional e-textile circuits [38] including enabling the quick removal of the BBC micro:bit from textiles, and creating solid electrical connections resulting in less time debugging compared to our prior workshops. Furthermore, our work recognizes the role of physical and material characteristics in hardware design. Previously Buechley et al. [9] tailored the LilyPad specifically for e-textile applications: their design deviated from the conventional square-shaped circuit boards, adopting a circular shape which spaced the plated through holes out aiding in sewing, and avoided awkward corners adversely impacting the fabric's drape (the way it hangs on a body in response to movement) [9]. The “SewSimple” design contributes to this discussion by demonstrating the advantages of these design decisions. Lastly, we present design recommendations for both hardware design and educational practices and will use these insights to inform our subsequent hardware iterations and our continuous contributions to supporting children's computational making development.

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8 Appendices

8.1 Participant Information

Table 2: Participant ID, Group ID, and Gender

ID	Group ID	Gender	ID	Group ID	Gender
P1	One	Female	P7	Six	Female
P2	One	Male	P8	Five	Male
P3	Two	Male	P9	Two	Male
P4	Four	Male	P10	Three	Male
P5	Five	Female	P11	Three	Female
P6	Six	Female	P12	Four	Male

8.2 Workshop Materials



Figure 5: Screenshots of selected instructional materials are part of a scenario-based design [42]. The first two images in the top row introduce background information about a new student named “Ming.” The second row contains two images that briefly describe “Ming’s” personal interests to inspire design.

8.3 “SewSimple” Design

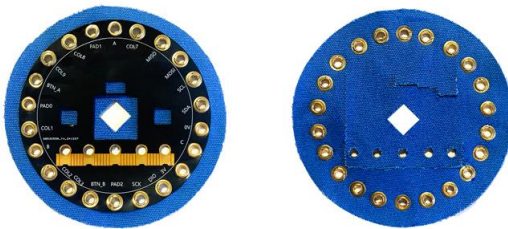


Figure 6: “SewSimple” board front and back.

8.4 Interview Questions

- When you first saw and touch the product, what was your immediate impression of it? Why do you feel that? (why so good/bad about it?)
- Can you tell me about the first thing you did with “SewSimple”? Did you need any extra help learning to use it?
- What did you create? How did you use “SewSimple” for your project? Can you walk me through it?
- What was the hardest/most frustrating part of using it? Why was it frustrating? How did you get around that issue? (or not, why)